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(54) **BUFFERING CLIENT FOR SYSTEM CONNECTION DETAILS**

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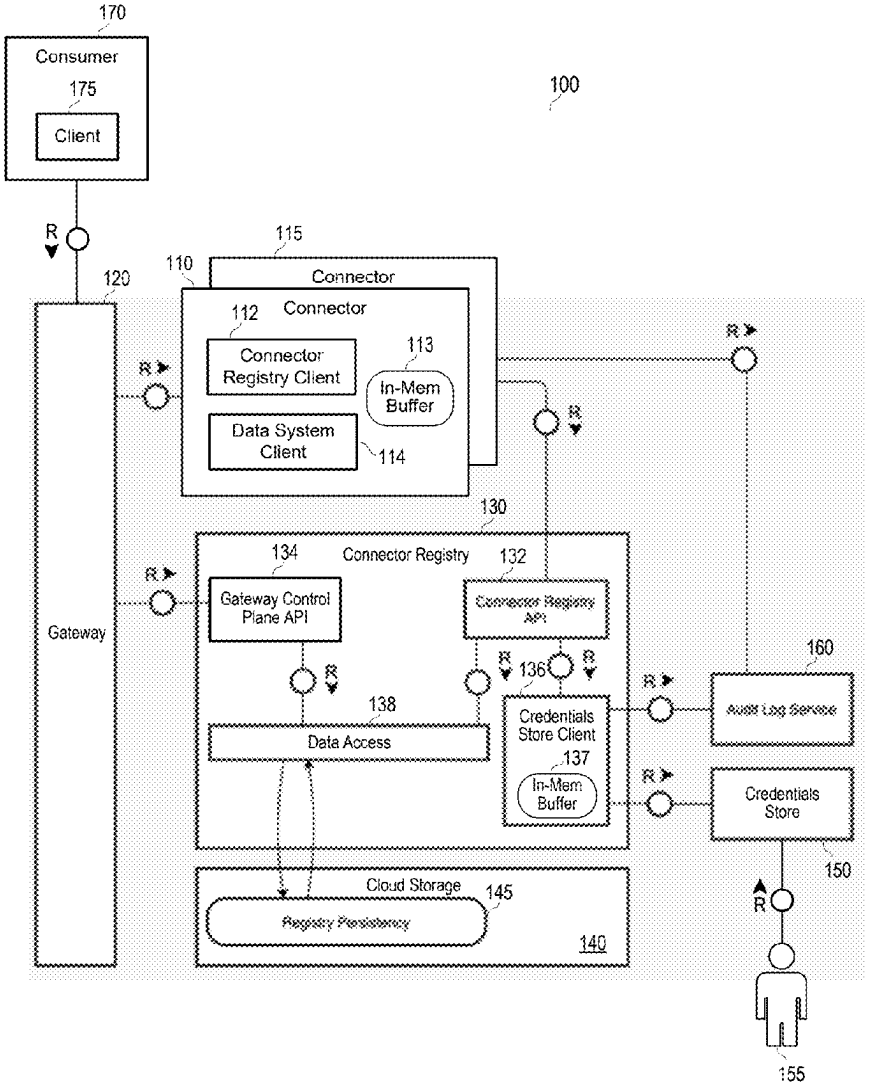
(57) **ABSTRACT**

Systems and methods include reception of a first request from a first instance of a first connector service, the first request specifying a first connection identifier, determination that first connection details associated with the first connection identifier are stored in a local memory buffer, transmission of the first connection details from the local memory buffer to the first instance in response to the determination that the first connection details are stored in the volatile memory buffer, and reception of a notification indicating that the first connection details have changed. In response to the notification, the notification is stored in a persistent storage, the first connection details stored in the local memory buffer are invalidated, and an instruction is transmitted to the first instance of the first connector service to invalidate the first connection details stored at the first instance.



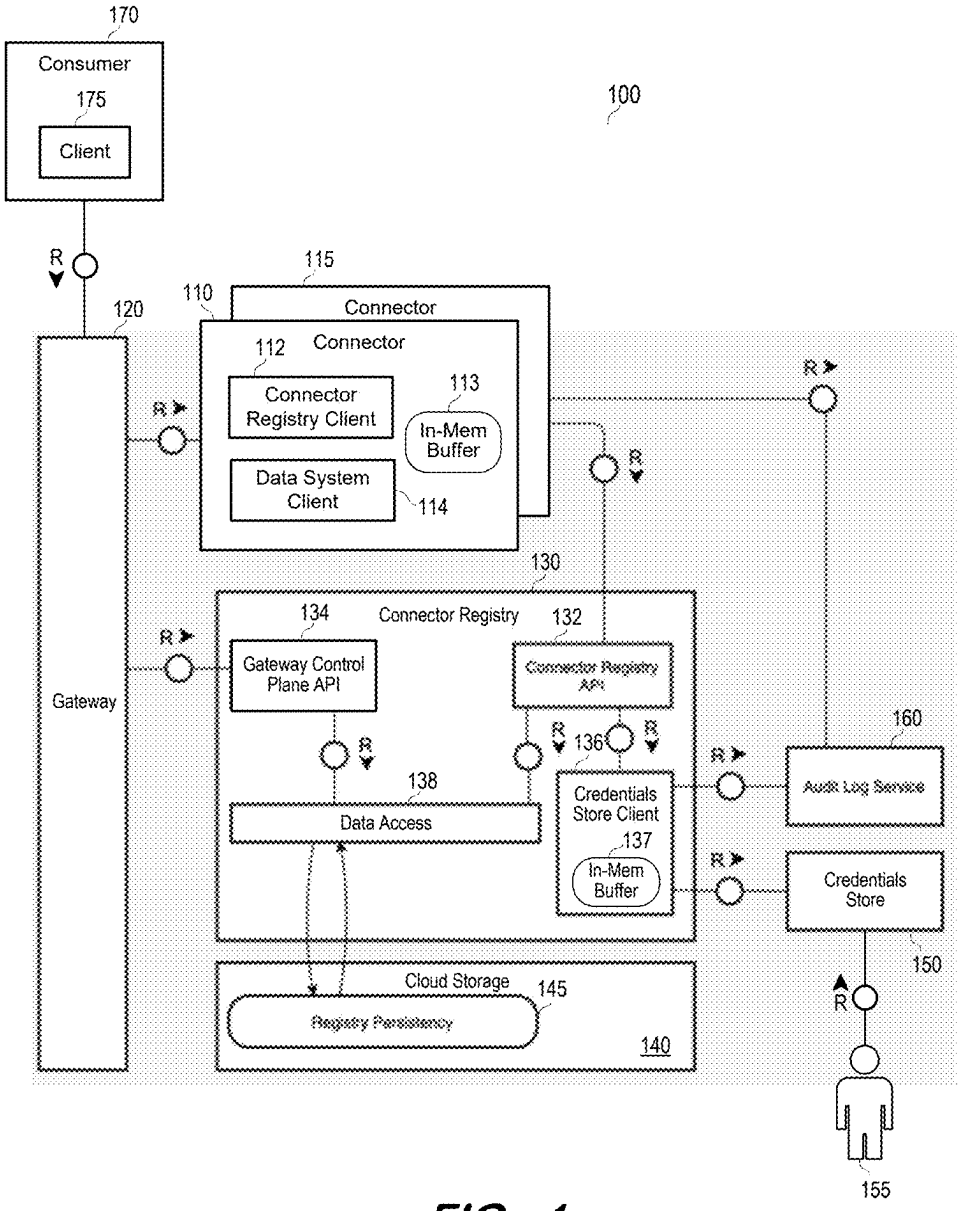

170
Consumer
175
Client
100
R
115
110
Connector
120
112
Connector
Connector Registry Client
113
In-Mem Buffer
Data System Client
114
130
134
Connector Registry
132
Gateway Control Plane API
Connector Registry API
136
138
Data Access
Credentials Store Client
137
In-Mem Buffer
160
Audit Log Service
Credentials Store
150
Gateway
Cloud Storage
145
Registry Persistency
140
155

FIG. 1

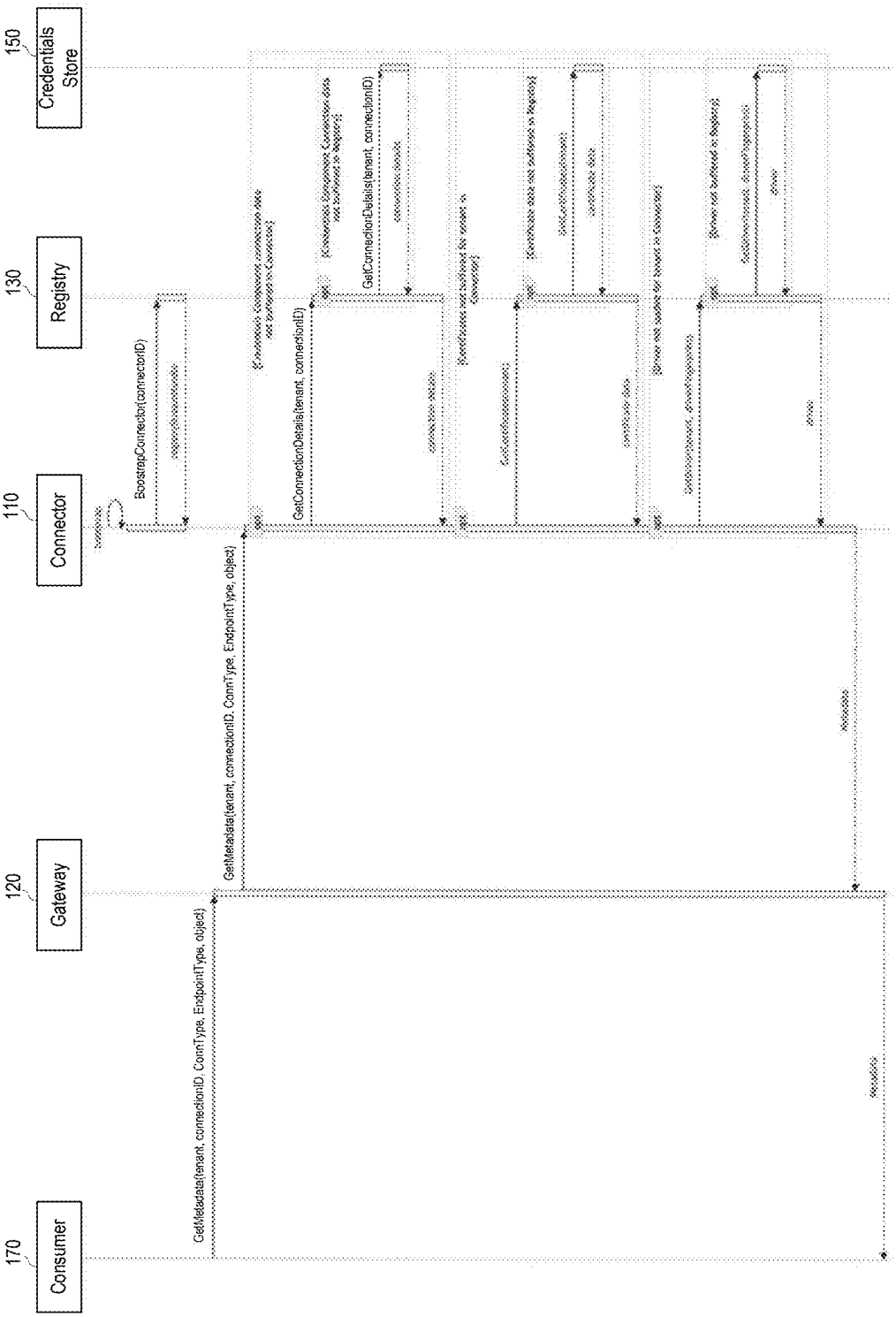

150
Credentials Store
130
Registry
110
Connector
120
Gateway
170
Consumer
BoostrapConnector(connectorID)
GetConnectionDetails(tenant, connectionID)
GetMetadata(tenant, connectionID, ConnType, EndpointType, object)
GetMetadata(tenant, connectionID, ConnType, EndpointType, object)

FIG. 2

310

CredentialsChangeLog			
timestamp	date_added	PK	can be drawn during insert directly from database
char36	tenant	PK	
char36	entity_id	PK	
short	entity_type		connection|certificate|endpoint|credential
short	operation		modification|deletion
char36	connection_id		if any associated to the change

FIG. 3

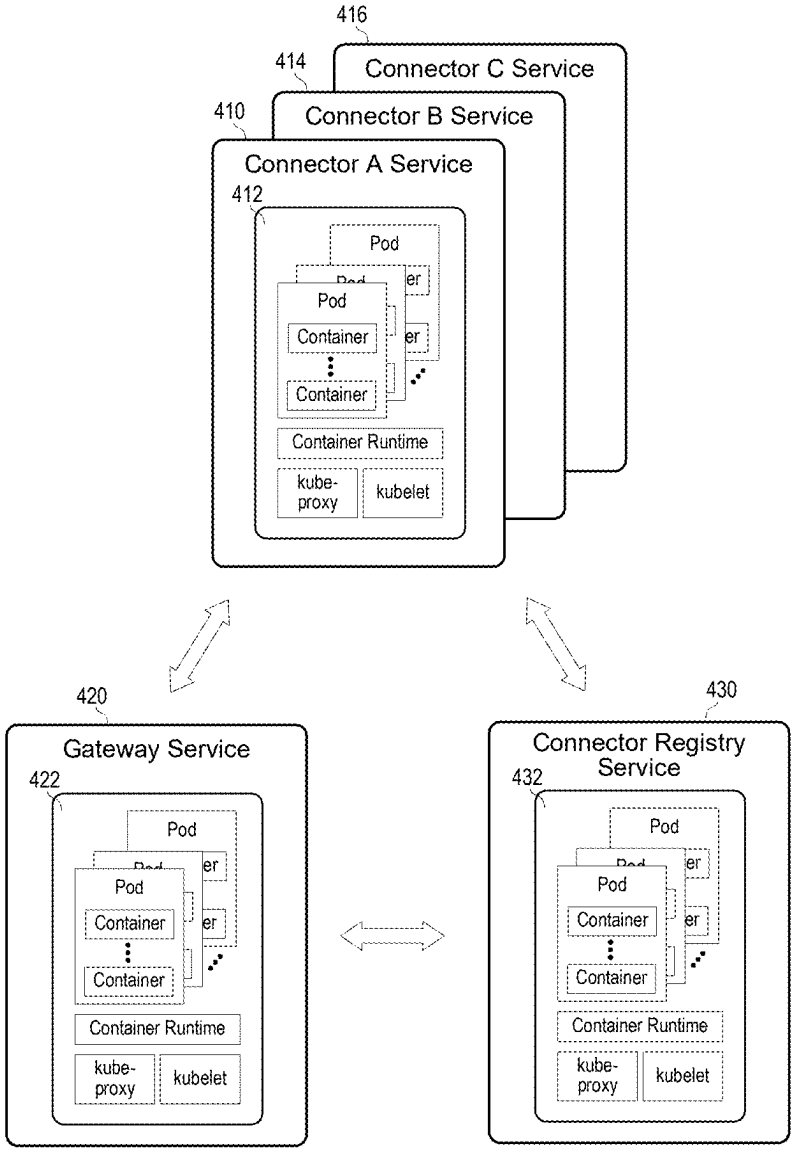

416
414
Connector C Service
410
Connector B Service
Connector A Service
412
Pod
Pod
Container
Container
Container Runtime
kube-proxy
kubelet
420
Gateway Service
422
Pod
Pod
Container
Container
Container Runtime
kube-proxy
kubelet
430
Connector Registry Service
432
Pod
Pod
Container
Container
Container Runtime
kube-proxy
kubelet

FIG. 4

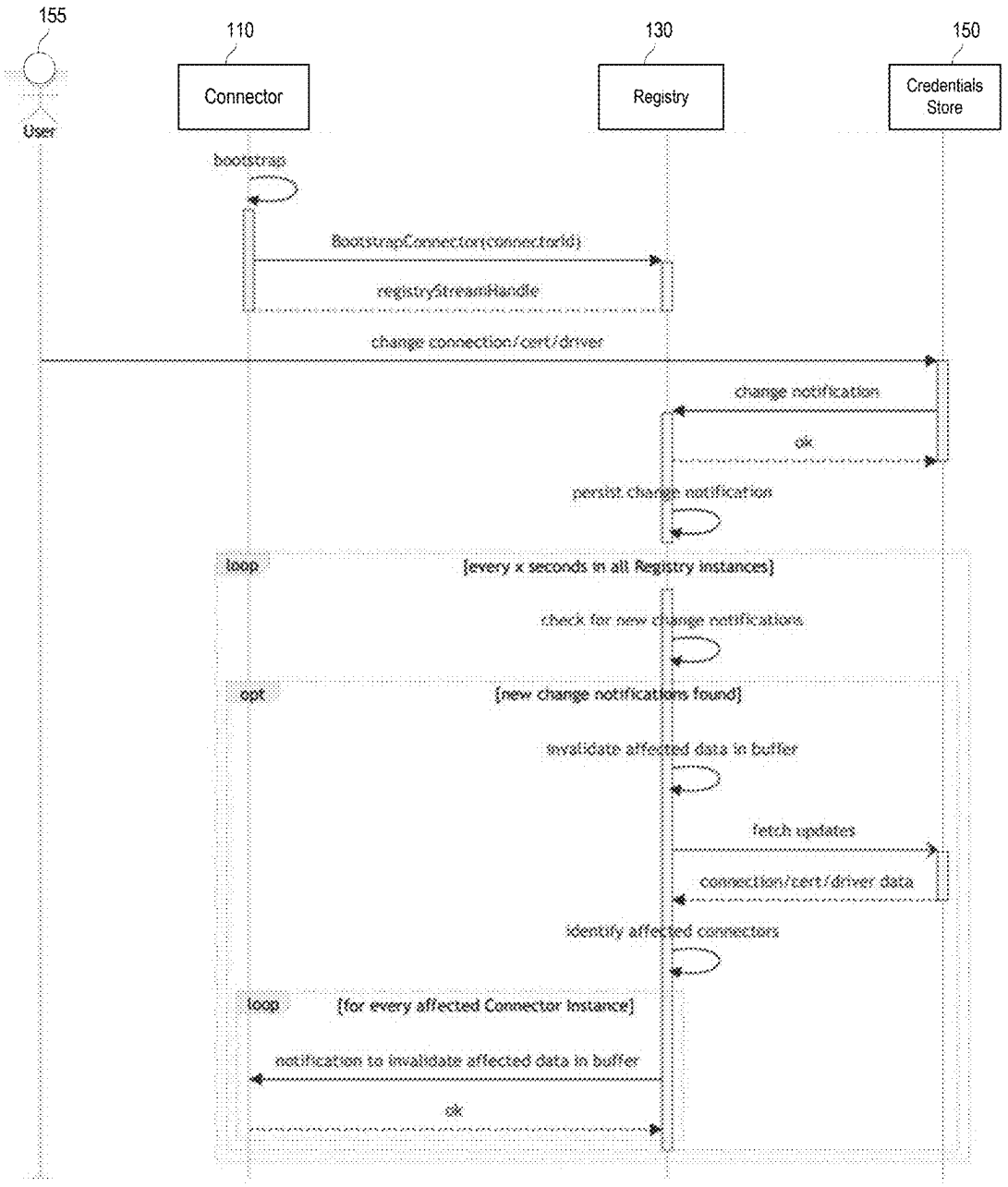

155
User
110
Connector
130
Registry
150
Credentials Store
bootstrap
BootstrapConnector(connectorId)
registryStreamHandle
change connection/cert/driver
change notification
ok
persist change notification
loop
{every x seconds in all Registry instances}
check for new change notifications
opt
{new change notifications found}
invalidate affected data in buffer
fetch updates
connection/cert/driver data
identify affected connectors
loop
{for every affected Connector instance}
notification to invalidate affected data in buffer
ok

FIG. 5

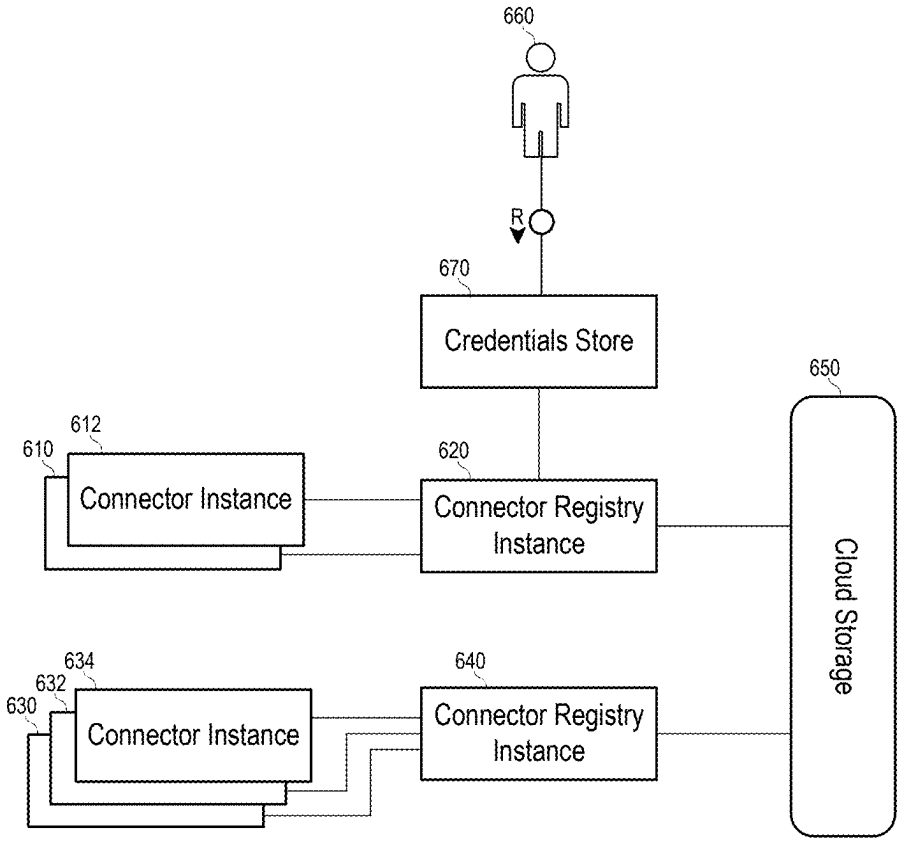

660
R
670
Credentials Store
650
612
610
Connector Instance
620
Connector Registry Instance
Cloud Storage
634
632
630
Connector Instance
640
Connector Registry Instance

FIG. 6

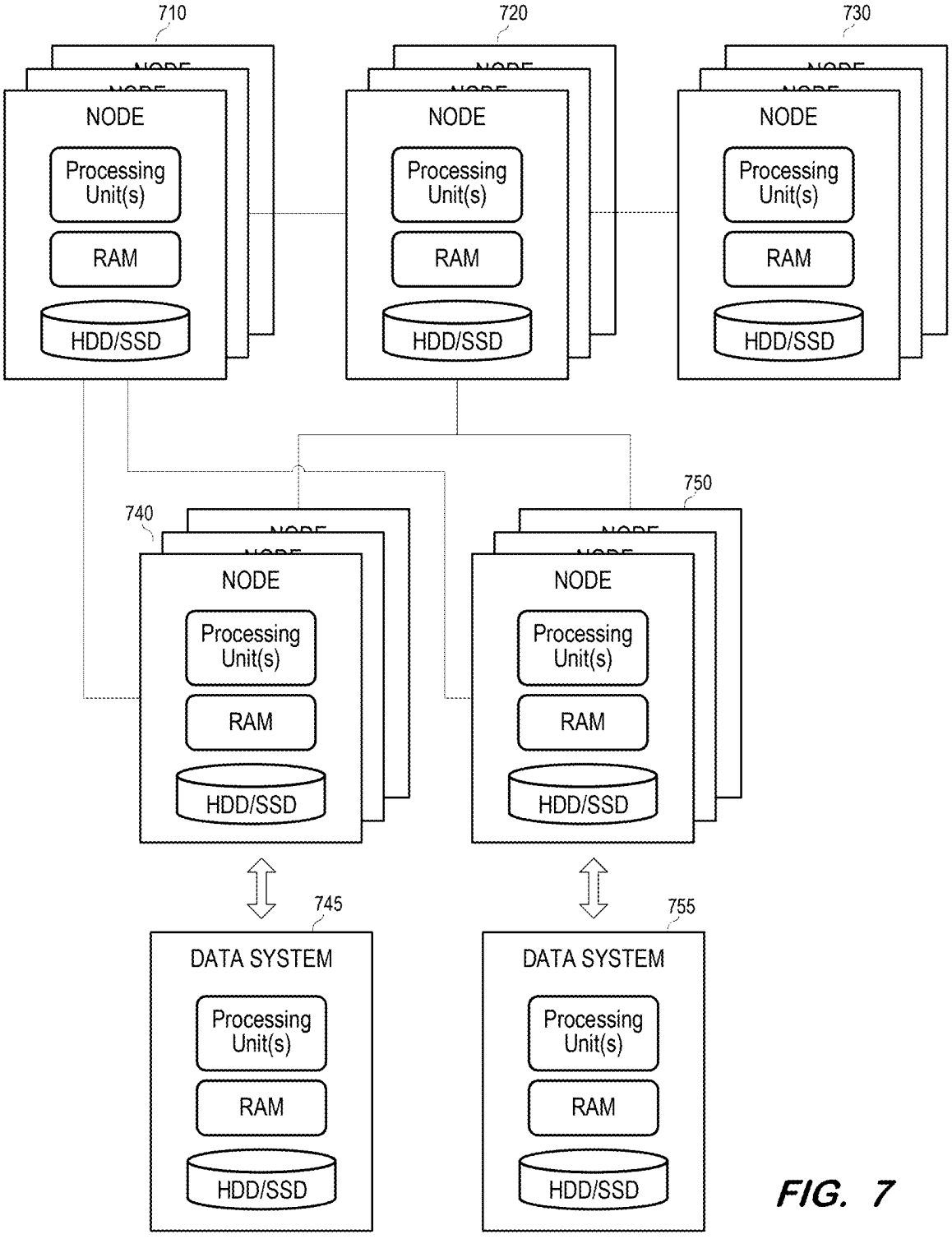

710
720
730
NODE
Processing Unit(s)
RAM
HDD/SSD
740
750
745
755
DATA SYSTEM

FIG. 7

BUFFERING CLIENT FOR SYSTEM CONNECTION DETAILS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/560,189, filed Mar. 1, 2024, under 35 U.S.C. § 119(e). The contents of U.S. Provisional Patent Application No. 63/560,189 are incorporated herein for all purposes.

BACKGROUND

[0002] Modern organizations generate, process and store vast amounts of data. This data may be stored in various data stores and accessed by various applications. Different data stores are differently suited for storing different types of data. Moreover, a given application may use different types of data associated with different requirements.

[0003] The particular data stores in which data of an application are stored may be selected based on many factors, including but not limited to a desired redundancy, a desired accessibility, a desired processing speed, and a desired security protocol. Due to these considerations, application data may be stored across multiple databases, file storages, file systems, and external data sources such as third-party vendors, cloud services, and cloud databases.

[0004] The diversity of data stores increases the complexity of data integration. For example, the more types of data stores which are used within an application landscape, the more difficult it is to seamlessly pull and push data and metadata from the data stores. Problems arise at least due to the difficulty in properly configuring the application to connect to and interact with the data stores. In addition, existing middleware for routing communication between an application and its data stores suffers from poor scalability and extensibility.

[0005] Connection to a data store typically requires corresponding user credentials and network details. This information may change from time-to-time and may also be subject to strict security protocols. For example, persistence of user credentials may be permitted only in secure data stores which conform to particular access and encryption standards. These freshness and data security issues pose further obstacles to seamless and efficient integration of data stores within an application landscape.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates a framework for connecting to data systems according to some embodiments.

[0007] FIG. **2** is a sequence diagram illustrating acquisition of connection details by a connector within a framework according to some embodiments.

[0008] FIG. **3** is an entity relationship diagram related to connector credentials change information according to some embodiments.

[0009] FIG. **4** illustrates hardware implementations of framework services according to some embodiments.

[0010] FIG. **5** is a sequence diagram illustrating invalidation of connection details in buffers of registry instances and connector instances according to some embodiments.

[0011] FIG. **6** illustrates invalidation of connection details in buffers of registry instances and connector instances according to some embodiments.

[0012] FIG. **7** illustrates a hardware implementation of a cluster-implemented framework according to some embodiments.

DETAILED DESCRIPTION

[0013] Embodiments may provide scalable, resilient, observable, extensible, and high-performance connectivity with disparate data systems.

[0014] FIG. **1** illustrates landscape **100** according to some embodiments. Landscape **100** may comprise any number of hardware and software components which provide functionality to one or more users (not shown). The components may be implemented using any suitable combinations of computing hardware and/or software that are or become known. Such combinations may include on-premise servers, cloud-based servers, and/or elastically-allocated virtual machines. In some embodiments, two or more components are implemented by a single computing device.

[0015] Landscape **100** comprises connectors **110** and **115**, which provide integration with data and metadata stored by data systems of different connection and endpoint types. Examples of connection types include but are not limited to ABAP, Google BigQuery, and S4/HANA Cloud, and examples of endpoint types include but are not limited to Web Socket RFC (WSRFC), REST, and Content Distribution Internetworking (CDI). Gateway **120** receives an incoming request for access to a data system of a particular connection type and endpoint type and routes the incoming request to a connector corresponding to the connection type, endpoint type and tenant. Connector registry **130** provides routing information to gateway **120** and connection details to deployed connectors **110** and **115**. Connection details may include information needed to connect to a data system, such as hostname, port, username, password, certificates, etc. Connector registry **130** may also manage connector registration, cache connection details in in-memory buffer **137**, and notify connectors **110**, **115** of connection detail updates and connection invalidations.

[0016] Embodiments may implement any number of connectors, each of which is associated with one or more respective connection types, endpoint types and tenants. A connector implements logic for reading and writing data and metadata from and to a data system instance of the connection type and endpoint type with which the connector is associated. Each connector **110**, **115** implements and exposes the same well-defined and parametrizable connector API which may be called by a consumer to interact with a data system of the connection type and endpoint type associated with the connector.

[0017] Consumer **170** may comprise a computing device which uses client component **175** to communicate with gateway **120** via the connector API. Client component **175** may encapsulate the communication and actions between consumer **170** and gateway **120**. Client component **175** may be tasked with managing gRPC connections, checking the backend's and connectors' health status, security, resilience, and other aspects.

[0018] For example, via gateway **120**, connector **110** may receive API calls from consumer **170** to read and write data, read and write metadata, and modify data definitions. Data system client **114** may convert the requests from a data model of connectors **110**, **115** to a data model of a connection type and endpoint type with which connector **110** is associated and transmit the requests to a data system using

connection details received from connector registry **130**. Similarly, data system client **114** may receive responses from a data system and convert the responses from the data model of the data system to the data model of connectors **110**, **115**.

[0019] A connector may provide consumer **170** with information specifying its characteristics and capabilities. For example, consumer **170** may transmit a request to gateway **120** requesting such capabilities associated with a particular connector ID. Connectors **110**, **115** may conform to a capability model which obliges a consumer to send commands to read or write data within the set of operations delimited by the connector's capabilities,

[0020] According to some embodiments, each connector **110**, **115** is a separate microservice that exposes and implements the connector API. Each microservice (i.e., connector) may be provided by a plurality of pods executing connector instances in one or more nodes of a container orchestration system as is known in the art. Landscape **100** may be a cloud-native and decoupled architecture in which gateway **120** and connector registry **130** each also comprise separate microservices implemented by a respective one or more nodes. Consequently and advantageously, each of connectors **110**, **115**, gateway **120** and connector registry **130** may scale elastically (e.g., by adding/deleting pods and/or nodes) according to their respective workloads. All microservices described herein may communicate with one another and with other unshown microservices using lightweight network communication mechanisms such as a resource API via Hyper Text Transfer Protocol (HTTP) request-response messages, but embodiments are not limited thereto.

[0021] According to some embodiments, landscape **100** uses gRPC for communication among the components, Protocol Buffers as an interface description language for message, service, and API definitions, and a data wire format such as Apache Arrow. Connectors **110**, **115**, gateway **120** and registry **130** may execute on an application server that runs and operates applications on Kubernetes.

[0022] Connector registry **130** comprises connector registry API **132**, gateway control plane API **134**, and credentials store client **136**. Connector registry API **132** may comprise a bi-directional gRPC API. During bootstrapping, connector registry client **112** of connector **110** invokes connector registry API **132** to register its information with connector registry **130**. The information may include, for example, connector ID, connector version, configuration, connection type, endpoint type, and supported tenants.

[0023] Connector registry API **132** calls data access object **138** to persist the connector-specific information in registry persistency **145** of cloud storage **140**. Registry persistency **145** may synchronize with the infrastructure layer of landscape **100** to also maintain a list of the available instances (e.g., running pods) for each registered connector.

[0024] Connector registry API **132** provides RPCs for connectors to retrieve connection details from credentials store client **136**. Credentials store client **136** connects to credentials store **150** to obtain connection details such as hostname, port, credentials associated with a connection ID. Connection details may also include certificate-related information as will be described below.

[0025] Due to the sensitivity of connection details, credentials store **150** may comprise a secure persistent storage system. User **155** (e.g., a tenant administrator) accesses credentials store **150** to store connection details for data systems to which the user has access. For example, each data system (and the connection details thereof) may be associated with a connection ID within credentials store **150**.

[0026] Credentials store client **136** includes local in-memory (i.e., volatile) buffer **137** to buffer connection details retrieved from credentials store **150**. Such buffering may avoid unnecessary calls to credentials store **150** and provide resilience in case of communication issues with credentials store **150**. Similarly, connector **110** also includes local volatile in-memory buffer **113** to buffer connection details received from connector registry **130** to avoid unnecessary calls to connector registry **130** and provide resilience in case of communication issues with connector registry **130**. Storage of the connection details in an in-memory buffer provides increased security in comparison to storage of the connection details in a persistent store.

[0027] In one example, connector **110** receives a request from gateway **120** including a connection ID and determines that connection details for the connection ID are not stored in buffer **113** (or that the connection details therein are for some reason invalid). Connector registry client **112** then requests connection details of connector **110** from connector registry **130**. In response, credentials store client **136** may determine that the requested details are not stored in buffer **137**. Credentials store client **136** requests the connection details from credentials store **150**, receives and caches the connection details in buffer **137**, and returns the connection details to connector **110** via API **132**. Connector **110** may then store the connection details in buffer **113** and use the connection details to forward a request to its associated data system.

[0028] Connector **110** may use the connection details stored in buffer **113** to respond to subsequent requests from gateway **120**. Similarly, credentials store client **136** may use the connection details stored in buffer **137** to respond to subsequent requests for connection details from connectors of the same connection type and endpoint type as connector **110**. If credentials store client **136** uses the connection details stored in buffer **137** to respond to a subsequent request for connection details, credentials store client **136** may forward an audit record to audit log service **160** to fulfill security and auditing requirements.

[0029] As will be described below, embodiments provide systems to ensure consistency of buffered connection details among the various simultaneously-executing instances of connector registry **130** and connectors **110**, **115**. Ensuring such consistency ensuring that connection details which are stored in instances of buffers **113** and **137** are not used by a connector if the connection details have been previously invalidated at credentials store **150**.

[0030] Gateway control plane API **134** provides routing information to gateway **120**. Gateway **120** caches the routing information and uses the routing information to route incoming requests to appropriate connectors. Gateway **120** routes incoming requests to connectors based on connection information (e.g., connection type, endpoint type, tenant) specified by the requests. Routing may be based on other types of information specified in a request. Gateway **120** may comprise an L7 load balancer that supports HTTP/2 and connections multiplexing, which allows the reuse of the same TCP socket for multiple HTTP connections.

[0031] Gateway **120** can survive outages of gateway control plane API **134** using cached routing information, at least until the routing information stored in registry persistency

145 changes. Routing information stored in registry persistency **145** may change due to asynchronous registration or de-registration of connectors. To keep the routing information of gateway **120** up-to-date, gateway **120** may maintain a live gRPC stream to gateway control plane API **134**. Accordingly, gateway control plane API **134** may regularly pull routing information from registry persistency **145** of cloud storage **140** via data access object **138** and determine whether any of the routing information has changed. If so, all current routing information stored in registry persistency **145** (or, in some embodiments, only the changed routing information) is transmitted to gateway **120**. According to some embodiments, prior to transmission of the routing information stored gateway control plane API **134** translates routing information entities corresponding to the connector data model into entities (e.g., configurations, actions and matchers) of the data model of gateway **120**.

[0032] As mentioned above, gateway **120** may be implemented using multiple instances. When a new instance is deployed, the instance calls gateway control plane API **134** to create a gRPC bi-directional stream and thereafter calls the gateway control plane API **134** to retrieve a routing configuration including all current routing information from registry persistency **145**.

[0033] FIG. **2** is a sequence diagram illustrating acquisition of connection details by a connector within a framework according to some embodiments. Process **200** and the other processes described herein may be performed using any suitable combination of hardware and software. Program code embodying these processes may be stored by any non-transitory tangible medium, including a fixed disk, a volatile or non-volatile random-access memory, a DVD, a Flash drive, or a magnetic tape, and executed by any number of processing units, including but not limited to processors, processor cores, and processor threads. Such processors, processor cores, and processor threads may be implemented by a virtual machine provisioned in a cloud-based architecture.

[0034] FIG. **2** will be described with respect to the components of landscape **100**, but embodiments are not limited thereto. During bootstrap of connector **110**, connector registry client **112** of connector **110** requests a connection with connector registry **130** via connector registry API **132** to open a long-lived gRPC channel associated with a connector ID of connector **110**. Registry **130** responds with a stream handle of the channel, which acts as a corridor for actions to be executed during bootstrap and across the life of connector **110** to allow registration of connector **110**, transfer of connection details from registry **130** to connector **110**, and determination of the status (e.g., running, terminated) of connector **110**.

[0035] During registration, connector **110** uses the corridor to call connector registry **130** and provides its connector information. The connector information may include a connector ID, the connection type and endpoint type to which connector **110** provides connectivity, the associated tenant (if not multi-tenant), a connector version, the endpoint of the registering connector instance, and any remaining configuration information needed for operation of connector **110** and routing of incoming requests. For example, the connector information may specify that connector **110** requires one or more drivers to communicate with its data system. In this case, the connector information may include fingerprints of the one or more drivers.

[0036] Connector registry **130** may translate the received connector information into native entities of a connector registry persistency data model. The translated connector information is persisted in cloud storage **140**. Next, connector registry **130** reads all the persisted connector information from cloud storage **140** and converts the entities of the connector information to a data model of gateway **120**. Connector registry **130** sends the converted connector information to gateway **120** to update its internal routing configuration. Accordingly, gateway **120** includes a routing configuration which reflects the current state of the connector information stored in persistency **145** of cloud storage **140**.

[0037] Returning to FIG. **2**, consumer **170** transmits a request to get metadata of a data system to gateway **120**. Consumer **170** may initially send a request to create a channel to gateway **120** via client component **175**, and the request to get metadata is then sent over the created channel. Although the example of FIG. **2** illustrates a request to get metadata, the request may comprise a request to read data/metadata, a request to write data, or a Data Description Language request according to some embodiments.

[0038] The request sent from consumer **170** includes, e.g., a tenant, a connection ID. a connection type, an endpoint type, and an object identifier. The connection ID is intended to correspond to a connection ID for which connection details were previously stored by the tenant in credentials store **150**. Accordingly, the request need not specify any connection details for connecting to the target data system.

[0039] Based on the received connection information and its stored routing information, gateway **120** routes the request to corresponding connector **110**. After receiving the request, connector **110** determines whether connection details corresponding to the connection ID of the request are stored in in-memory buffer **113**. If not, connector **110** calls registry **130** (e.g., using a connector registry client as described above) to request the connection details.

[0040] Connector registry **130** determines whether connection details corresponding to the connection ID and tenant are stored in in-memory buffer **137**. If so, registry **130** sends the connection details to connector **110** for storage in buffer **113**. If the connection details are not in in-memory buffer **137**, credentials store client **136** of connector registry **130** calls credentials store **150** to request the connection details, stores the connection details in buffer **137**, and sends the connection details back to connector **110** for storage in buffer **113**.

[0041] The connection details received from registry **130** (or already stored in buffer **113**) may include a certificate fingerprint indicating that a certificate is required to access the data system. If so, and if the certificate is not already stored in in-memory buffer **113**, connector **110** requests the certificate corresponding to the certificate fingerprint and tenant from the registry from registry **130**. In response, connector registry **130** determines whether the requested certificate is stored in in-memory buffer **137**. If so, registry **130** returns the certificate to connector **110**. If not, connector registry **130** calls credentials store **150** to request the certificate, stores the certificate in buffer **137**, and returns the certificate to connector **110**. In either case, the returned certificate may be stored in buffer **113**.

[0042] As previously mentioned, connectors may require drivers to operate. The drivers might not be bundled into the connectors due to licensing or other restrictions. Users can

therefore upload their licensed drivers for the tenant to credentials component **150**. When connector **110**, for example, receives a request, it looks for the tenant information and checks whether its driver is stored in in-memory buffer **113**. If not, connector **110** contacts the registry **130** for the driver for that tenant.

[0043] Registry **130** determines whether the driver is stored in in-memory buffer **137** and, if so, returns the driver to connector **110**. If the driver is not stored in in-memory buffer **137**, registry **130** calls credentials store **150** to determine whether the driver is stored in credentials store **150**. If so, registry **130** calls credentials store **150** to retrieve the driver and returns the driver to connector **110**. Connector **110** may store the returned driver in buffer **113**.

[0044] Connector **110** then uses the connection details, certificate (if needed) and driver (if needed) to transmit the request for metadata to the target data system (not shown). Connector **110** receives the result and returns the result to consumer **170** via gateway **120**. If the connection details, certificate, or driver are not available in buffers **113**, **137** or credentials store **150**, an error is returned to connector **110** and may be forwarded to consumer **170**.

[0045] In some embodiments, buffering the driver, certificate, and connection details in connector registry **130** advantageously provides faster response times and resilience against an outage of credentials store **150**. Since drivers are large (e.g., tens of MB), tenant-specific, and/or subject to licensing, some embodiments do not buffer drivers and instead load drivers from credentials store **150** into corresponding connectors at startup.

[0046] FIG. **4** illustrates implementations of connectors **410**, **414** and **416** gateway **420** and connector registry **430** according to some embodiments. Each of services **410**, **414**, **416**, **420** and **430** may be implemented by a plurality of pods of a container orchestration platform such as but not limited to Kubernetes. As such, each service may include a node executing several pods, in which each pod executes a separate instance of the service. Each node **412**, **422** and **432** is a virtual or physical machine including kubelet for node and container management, kube-proxy for a network proxy, and a container runtime (e.g., Docker) to run container.

[0047] Although FIG. **4** illustrates one node **412**, **422** and **432** per service, any service may include any number of nodes, each executing its own set of one or more pods. A master node (not shown) of each service may adjust the number of pods, the number of nodes and/or the computing resources of each node as desired. The adjustment may be based on expected workload or any other factors. For example, if an expected workload is greater than a first threshold, one or more additional pods are created. If the expected workload is less than a second threshold, one or more of pods are terminated.

[0048] Accordingly, each of connector services **410**, **414**, **416** may be implemented by several connector instances and connector registry service **430** may be implemented by several connector registry instances, in which all instances execute independently and in parallel. If a connector registry client of a connector instance registers its connector information using a connector registry API of a connector registry instance, all other connector registry instances will initially be unaware of the registration. Accordingly, to establish a consistent state of connector information throughout the landscape, each running connector registry instance may periodically read the connector information from the shared cloud storage and push the connector information to the gateway instance(s) to which it is connected.

[0049] As described above, some embodiments advantageously buffer connection details, certificates, and drivers in a connector registry and/or in connectors. If connection details, certificates, and/or drivers are changed or deleted in credentials store **150**, all connector registry instances and each connector instance which buffer the changed/deleted data should be notified.

[0050] This notification is accomplished in some embodiments by providing a callback for push notifications in credentials store **150**. Using this callback, one connector registry instance receives notifications of changes/deletions from credentials store **150**, and the notifications are propagated to all other connector registry instances. Each connector registry instance then sends the notifications to the connector instances which established a bidirectional connection with the connector registry instance during bootstrapping as described above.

[0051] FIG. **5** is a sequence diagram illustrating invalidation of connection details in buffers of registry instances and connector instances according to some embodiments. During bootstrap of connector **110**, a long-lived gRPC channel associated with a connector ID of connector **110** is opened with registry **130**. The channel is used to register information of connector **110** and to transfer connection details from registry **130** for use and/or buffering therein.

[0052] During operation, user **155** may communicate with credentials store **150** to change connection details, a driver and/or a certificate associated with a connection ID. Using the above-described callback, credentials store **150** transmits a notification to registry **130** notifying registry **130** of the change. Registry **130** acknowledges the notification and persists the notification to cloud storage **140**. The notification may identify invalidated connection details, drivers, and certificates per tenant. The notification may be timestamped to facilitate purging of stale change information from storage **140**. FIG. **3** illustrates attributes of a CredentialsChangeLog entity **310** which may be used as a notification and persisted in cloud storage **140** as described above according to some embodiments.

[0053] Each connector registry instance may be programmed in some embodiments to periodically (e.g., every 10 s) check the persistence for change notifications which was stored since a last check. Assuming that a registry instance identifies a change notification which was stored since its last check, the connector registry instance and invalidates (e.g., deletes from the buffer or marks as invalid) the identified changed connection details, certificates, and/or drivers.

[0054] Next, the connector registry instance fetches the changed connection details, certificates, and/or drivers from the persistence, certificates or drivers stored in its in-memory buffer have been changed. The connector registry instance identifies connector instances to which it is connected, and which are associated with the connection ID corresponding to the changes. The connector registry instance then instructs these connector instances to invalidate the changed data (i.e., connection details, certificate, and/or driver) if stored in their buffers. In some embodiments, the connector registry instance pushes the new data to corresponding connector instances with instructions to update their buffers with the new data.

[0055] FIG. **6** illustrates component instances for describing an example of the sequence diagram of FIG. **5**. User **660** changes connection details, a driver and/or a certificate stored in credentials store **670** in association with a connection ID. Credentials store client **150** transmits a notification of the change to connector registry instance **620**. Connector registry instance **620** persists the change notification to cloud storage **650**.

[0056] Soon thereafter, connector registry instance **620** checks cloud storage **650** for change notifications which were stored since a last check. Connector registry instance **620** determines that a change notification was stored since its last check and invalidates connection details, certificates, and/or drivers identified in the change notification and stored in its in-memory buffer.

[0057] Connector registry instance **620** then fetches the changed information from cloud storage **650**, identifies connector instances **610**, **612** to which it is connected via respective gRPC long-lived channels, and determines whether any of connector instances **610**, **612** are associated with the connection ID and/or tenant corresponding to the changes. Connector registry instance **620** instructs its connector instances associated with the connection ID and/or tenant to invalidate the changed data (i.e., connection details, certificate, and/or driver) if the prior version is stored in their buffers. In some embodiments, the connector registry instance pushes the new data to corresponding connector instances with instructions to update their buffers with the new data.

[0058] Similarly, and independently from connector registry instance **620**, connector registry instance **640** checks cloud storage **650** for change notifications which were stored since its last check and fetches the recently-stored change notifications from cloud storage **650**. Connector registry instance **640** invalidates any changed connection details, certificates, or drivers from its in-memory buffer, fetches the changes, and determines whether any of connector instances **630**, **632**, **634** with which it is connected are associated with the connection ID and/or tenant corresponding to the changes. Connector registry instance **640** instructs such ones of connector instances **630**, **632**, **634** to invalidate the changed data if the prior version of the data is stored in their buffers. In some embodiments, each connector registry instance updates its in-memory buffer with newly-changed data and/or pushes the newly-changed data to its connected connector instances with instructions to update their buffers with the data.

[0059] FIG. **7** illustrates a cloud-based deployment according to some embodiments. The illustrated components may comprise cloud-based resources residing in one or more public or private clouds providing self-service and immediate provisioning, autoscaling, security, compliance and identity management features.

[0060] Each illustrated node may comprise a physical or virtual machine of a container orchestration platform cluster. Nodes **710** may execute instances of a gateway while nodes **720** may execute instances of a connector registry. Nodes **730** may comprise a database service for storing connector-specific information. Nodes **740** may execute instances of a first connector and nodes **750** may execute instances of a second connector. The instances of the first connector may communicate with data system **745** and the instances of the second connector may communicate with data system **755**.

[0061] The foregoing diagrams represent logical architectures for describing processes according to some embodiments, and actual implementations may include more, or different components arranged in other manners. Other topologies may be used in conjunction with other embodiments. Moreover, each component or device described herein may be implemented by any number of devices in communication via any number of other public and/or private networks. Two or more of such computing devices may be located remote from one another and may communicate with one another via any known manner of networks and/or a dedicated connection. Each component or device may comprise any number of hardware and/or software elements suitable to provide the functions described herein as well as any other functions. For example, any computing device used in an implementation of a system according to some embodiments may include a processor to execute program code such that the computing device operates as described herein.

[0062] All systems and processes discussed herein may be embodied in program code stored on one or more non-transitory computer-readable media. Such media may include, for example, a hard disk, a DVD-ROM, a Flash drive, magnetic tape, and solid-state Random Access Memory (RAM) or Read Only Memory (ROM) storage units. Embodiments are therefore not limited to any specific combination of hardware and software.

[0063] Embodiments described herein are solely for the purpose of illustration. Those in the art will recognize other embodiments may be practiced with modifications and alterations to that described above.

What is claimed is:

1. A system comprising:

a memory storing program code;

a volatile memory buffer;

one or more processing units to execute the program code to cause the system to execute a first instance of a first connector registry service to:

receive a first request from a first instance of a first connector service, the first request specifying a first connection identifier;

determine that first connection details associated with the first connection identifier are stored in the volatile memory buffer;

in response to the determination that the first connection details are stored in the volatile memory buffer, transmit the first connection details from the volatile memory buffer to the first instance of the first connector service;

receive a notification indicating that the first connection details have changed;

and in response to the notification indicating that the first connection details have changed:

store the notification in a persistent storage; and

invalidate the first connection details stored in the volatile memory buffer.

2. The system of claim **1**, the one or more processing units to execute the program code to cause the system to execute the first instance of the first connector registry service to:

transmit an instruction to the first instance of the first connector service to invalidate the first connection details stored at the first instance of the first connector service.

3. The system of claim 1, the one or more processing units to execute the program code to cause the system to execute the first instance of the first connector registry service to:

receive a second request from a first instance of a second connector service, the second request specifying a second connection identifier;

determine that second connection details associated with the second connection identifier are not stored in the volatile memory buffer; and

in response to the determination that the second connection details are not stored in the volatile memory buffer:

request the second connection details from a secure data store;

receive the second connection details from the secure data store;

store the second connection details in the volatile memory buffer; and

transmit the second connection details to the first instance of the second connector service.

4. The system of claim 3, wherein the notification is received from the secure data store.

5. The system of claim 3, the one or more processing units to execute the program code to cause the system to execute the first instance of the first connector registry service to:

receive a second notification indicating that the second connection details have changed; and

in response to the second notification indicating that the second connection details have changed:

store the second notification in the persistent storage; and

transmit a second instruction to the first instance of the second connector service to invalidate the second connection details stored at the first instance of the second connector service.

6. The system of claim 5, wherein the notification and the second notification are received from the secure data store.

7. The system of claim 1, the one or more processing units to execute the program code to cause the system to execute a second instance of the first connector registry service to:

determine the first connection details stored in the persistent storage; and

in response to determination of the first connection details, transmit an instruction to a second instance of the first connector service to invalidate the first connection details stored at the second instance of the first connector service.

8. The system of claim 7, the one or more processing units to execute the program code to cause the system to execute a second instance of the first connector registry service to:

in response to determination of the notification, determine that a first instance of a second connector service with which the first connector registry service is in communication is not associated with the first connection details; and

in response to the determination that the first instance of the second connector service is not associated with the first connection details, do not transmit an instruction to the first instance of the second connector service to invalidate the first connection details.

9. A method comprising:

receiving, at a first instance of a first connector registry service, a first request from a first instance of a first connector service, the first request specifying a first connection identifier;

determining, at the first instance of the first connector registry service, that first connection details associated with the first connection identifier are stored in a local volatile memory buffer;

in response to determining that the first connection details are stored in the local volatile memory buffer, transmitting the first connection details from the first instance of the first connector registry service to the first instance of the first connector service;

receiving, at the first instance of the first connector registry service, a notification indicating that the first connection details have changed; and

in response to the notification indicating that the first connection details have changed:

storing the notification in a persistent storage; and

invalidating the first connection details stored in the local volatile memory buffer.

10. The method of claim 9, further comprising:

transmitting, from the first instance of the first connector registry service, an instruction to the first instance of the first connector service to invalidate the first connection details stored at the first instance of the first connector service.

11. The method of claim 9, further comprising:

receiving, at the first instance of the first connector registry service, a second request from a first instance of a second connector service, the second request specifying a second connection identifier;

determining, at the first instance of the first connector registry service, that second connection details associated with the second connection identifier are not stored in the local volatile memory buffer; and

in response to determining that the second connection details are not stored in the local volatile memory buffer:

requesting, by the first instance of the first connector registry service, the second connection details from a secure data store;

receiving, at the first instance of the first connector registry service, the second connection details from the secure data store;

storing, by the first instance of the first connector registry service, the second connection details in the local volatile memory buffer; and

transmitting, from the first instance of the first connector registry service, the second connection details to the first instance of the second connector service.

12. The method of claim 11, wherein the notification is received from the secure data store.

13. The method of claim 11, further comprising:

receiving at the first instance of the first connector registry service, a second notification indicating that the second connection details have changed; and

in response to the second notification indicating that the second connection details have changed:

storing, by the first instance of the first connector registry service, the second notification in the persistent storage; and

transmitting, from the first instance of the first connector registry service, a second instruction to the first instance of the second connector service to invalidate the second connection details stored at the first instance of the second connector service.

14. The method of claim **13**, wherein the notification and the second notification are received from the secure data store.

15. The method of claim **9**, further comprising:

determining, at a second instance of the first connector registry service, the notification stored in the persistent storage; and

in response to determination of the notification, transmitting, from the second instance of the first connector registry service, an instruction to a second instance of the first connector service to invalidate the first connection details stored at the second instance of the first connector service.

16. The method of claim **15**, further comprising:

in response to determining the notification, determining, at the second instance of the first connector registry service, that a first instance of a second connector service with which the first connector registry service is in communication is not associated with the first connection details; and

in response to determining that the first instance of the second connector service is not associated with the first connection details, not transmitting, from the second instance of the first connector registry service, an instruction to the first instance of the second connector service to invalidate the first connection details.

17. One or more non-transitory computer-readable media storing program code that, when executed by a computing system, cause the computing system to perform operations comprising:

receiving a first request from a first instance of a first connector service, the first request specifying a first connection identifier;

determining that first connection details associated with the first connection identifier are stored in a local memory buffer;

in response to the determination that the first connection details are stored in the volatile memory buffer, transmitting the first connection details from the local memory buffer to the first instance of the first connector service;

receiving a notification indicating that the first connection details have changed; and

in response to the notification indicating that the first connection details have changed:

storing the notification in a persistent storage;

invalidating the first connection details stored in the local memory buffer; and

transmitting an instruction to the first instance of the first connector service to invalidate the first connection details stored at the first instance of the first connector service.

18. The one or more non-transitory computer-readable media of claim **17**, the program code, when executed by a computing system, to cause the computing system to perform operations comprising:

receiving a second request from a first instance of a second connector service, the second request specifying a second connection identifier;

determining that second connection details associated with the second connection identifier are not stored in the local memory buffer; and

in response to determining that the second connection details are not stored in the local memory buffer:

requesting the second connection details from a secure data store;

receiving the second connection details from the secure data store;

storing the second connection details in the local memory buffer; and

transmitting the second connection details to the first instance of the second connector service.

19. The one or more non-transitory computer-readable media of claim **18**, the program code, when executed by a computing system, to cause the computing system to perform operations comprising:

receiving a second notification indicating that the second connection details have changed; and

in response to the second notification indicating that the second connection details have changed:

storing the second notification in the persistent storage; and

transmitting a second instruction to the first instance of the second connector service to invalidate the second connection details stored at the first instance of the second connector service.

20. The one or more non-transitory computer-readable media of claim **19**, the program code, when executed by a computing system, to cause the computing system to perform operations comprising:

determining the notification stored in the persistent storage; and

in response to determining the notification, transmit an instruction to a second instance of the first connector service to invalidate the first connection details stored at the second instance of the first connector service.

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